Chapter 1
Science and Measurement

Chemical Principles I

CHEM 1211K

Learning Objectives

1. Classify matter, distinguishing between pure substances and mixtures, elements and compounds, homogeneous and heterogeneous mixtures, etc.

2. Distinguish between physical and chemical changes and properties.

3. Outline the principles of the scientific approach to the study of molecules.

4. Report scientific measurements to reflect level of certainty (precision).

5. Work with significant digits, reporting calculated values with an appropriate level of precision.

6. Use conversion factors to convert between different units and apply dimensional analysis.

1.1 Classification of Matter |

- **Matter** is anything that has mass and takes up space.

- Matter is classified according to…
  - **State:** solid, liquid, or gas?
  - **Composition:** what kinds of substances compose it, and in what amounts?

- **Pure substances** contain only a single type of atom, molecule, or formula unit.
  - **Elements** cannot be broken down into simpler substances.
  - **Compounds** are composed of two or more elements in fixed and definite proportions

- **Mixtures** are composed of two or more different types of atoms or molecules combined in variable proportions.
  - **Heterogeneous** mixtures differ in composition at different points in the substance.
  - **Homogeneous** mixtures have the same composition at every point in the mixture.

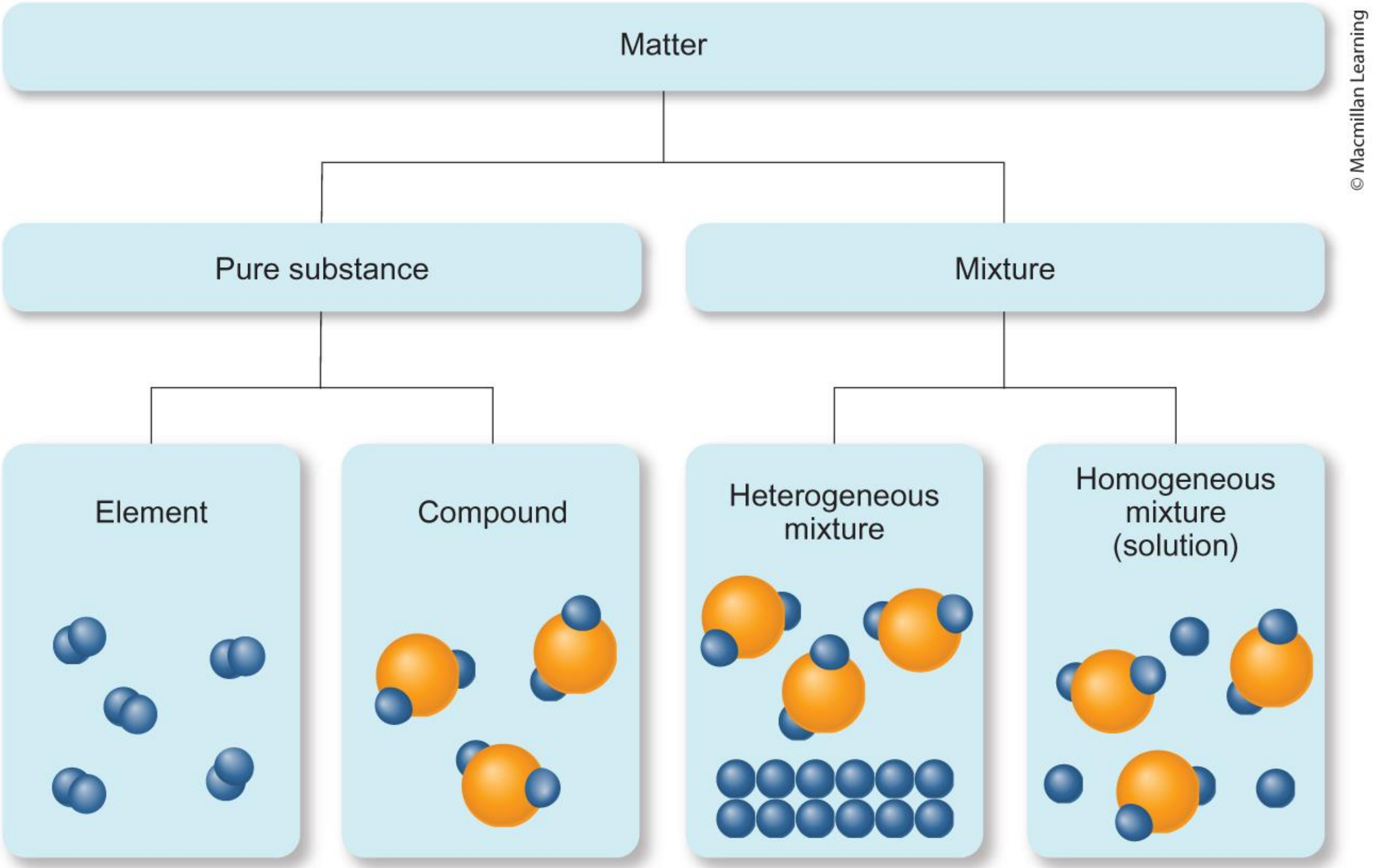
Matter
Pure substance
Mixture
Element
Compound
Heterogeneous mixture
Homogeneous mixture (solution)
© Macmillan Learning

Figure 1.2

1.2 Properties of Matter |

- **Properties** are characteristics by which we can identify or describe a substance.
  - Examples: color, odor, state of matter, temperature of a phase transition, chemical reactivity, etc.

- **Physical properties** involve processes or characteristics that do not affect the chemical identity of the substance.
  - Examples: color, odor, density, melting point, state of matter, viscosity, etc.

- **Chemical properties** involve processes or characteristics associated with a chemical change.
  - Examples: flammability, acidity, toxicity, corrosiveness, etc.

- **Extensive properties** have values that depend on the quantity of the sample
  - Examples: mass, volume, number of moles, etc.

- **Intensive properties** have the same value regardless of the quantity of the sample
  - Examples: boiling point, density, [anything] per gram, [anything] per mole, etc.

1.2 Properties of Matter | 2 of 2

- **Physical changes** alter the state or appearance of matter but do not affect its chemical composition.

Melting wax (a) and condensing water (b) are physical changes.

(a) (b)

Figure 1.18 (OpenStax Chemistry 2e)

- **Chemical changes (chemical reactions)** alter the chemical structure of a substance by breaking and/or forming chemical bonds.

Reaction of iron with oxygen (a) is a chemical change; chromium does not react with oxygen (b).

(a) (b)

Figure 1.19 (OpenStax Chemistry 2e)

- **Hypothesis**
  - A tentative interpretation or explanation of observations
  - Should be *falsifiable*, making predictions that can be supported or refuted by further observation

- **Experiments**
  - Highly controlled procedures designed to generate observations that can support or refute a hypothesis

- **Scientific theory**
  - A model for the way nature is that attempts to explain not merely what nature does, but why.
  - Often, theories predict behavior far beyond the observations or laws from which they were developed.
  - Example: Dalton's atomic theory proposed that matter is composed of small, indestructible particles (atoms) that rearrange during chemical changes such that the total amount of mass remains constant.

- **Scientific law**
  - A brief statement that summarizes past observations and predict future ones
  - Example: The *law of conservation of mass:* "In a chemical reaction, matter is neither created nor destroyed."
  - Subject to experiments which can add support to them or prove them wrong (like hypotheses)

- Laws are scientific observations that have always been seen to be true, but they are not explanations for why the observation is always that way.
- Theories are scientific explanations that so far have not been proven wrong.
- A theory and a law are two entirely different things.
- A hypothesis can become a theory, but a theory can never become a law.

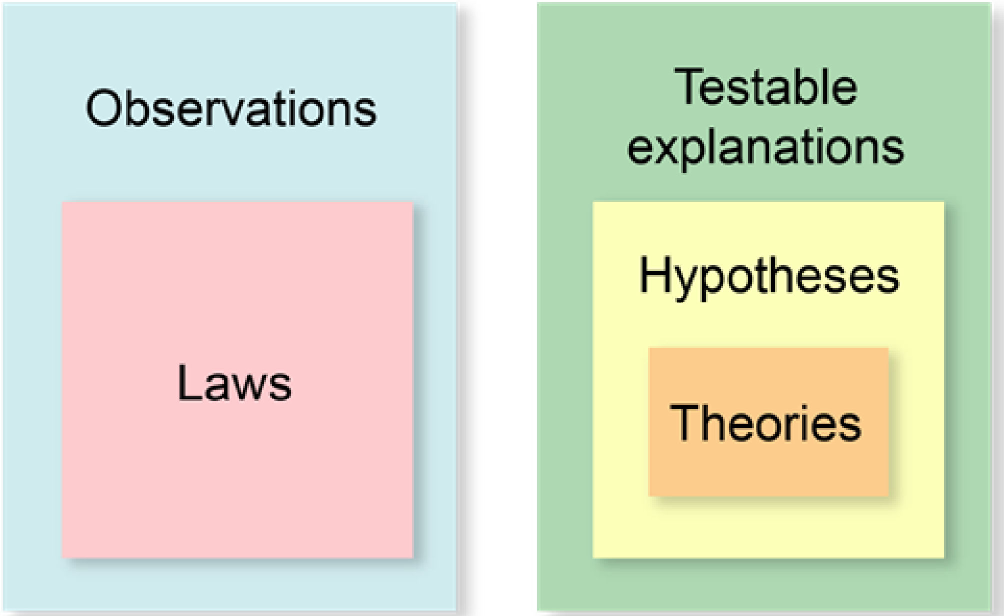

Figure 1.9

- Scientific knowledge is **empirical**—based on observation and experiment.

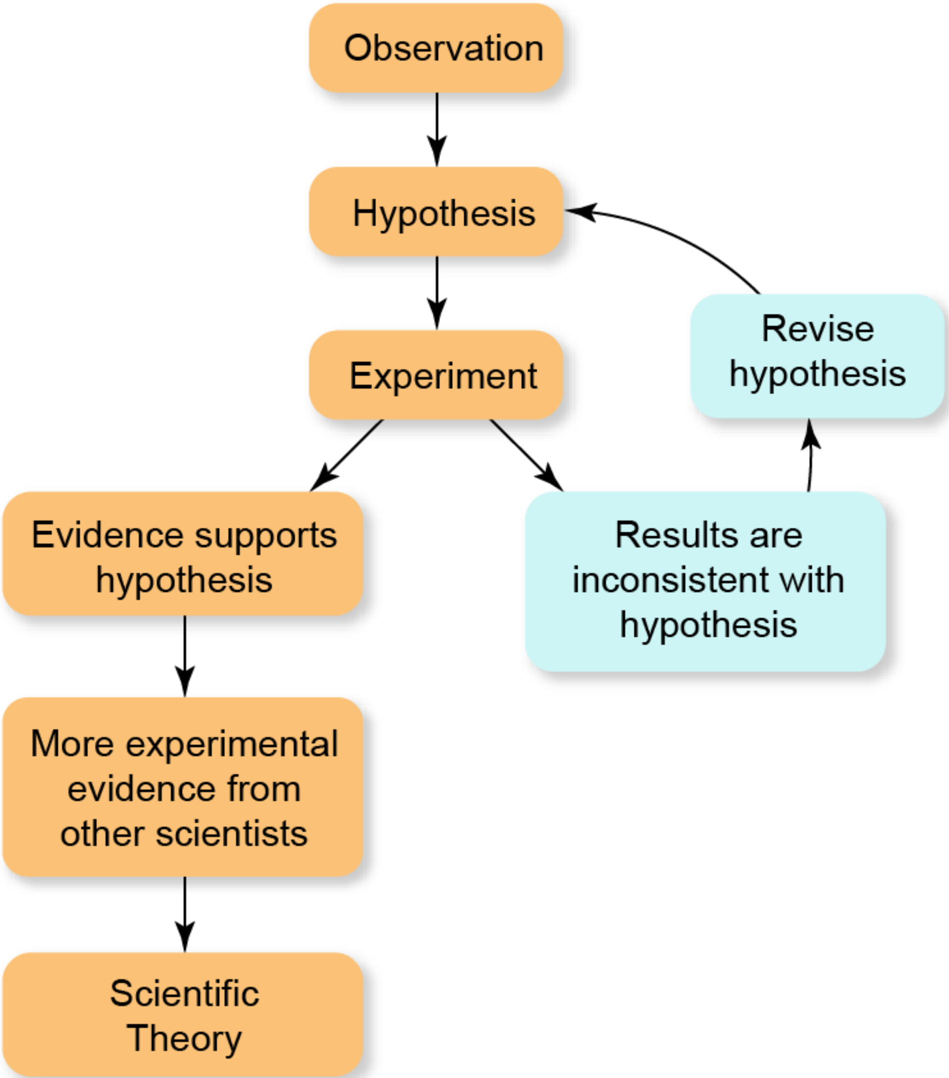

Figure 1.8

- Scientists use the **International System of Units (SI)**, which is based on the metric system.

SI Base Units		
Quantity	**Unit**	**Symbol**
Length	meter	m
Mass	kilogram	kg
Time	second	s
Temperature	Kelvin	K
Amount of substance	mole	mol
Electric current	Ampere	A
Luminous intensity	candela	cd

- **Scientific notation** allows us to express very large or very small quantities in a compact way using positive or negative exponents.

- The metric system uses standardized prefixes to represent multipliers on units

SI Prefixes		
Prefix	**Abbreviation**	**Meaning**
tera-	T	$\times 10^{12}$
giga-	G	$\times 10^{9}$
mega-	M	$\times 10^{6}$
kilo-	k	$\times 10^{3}$
deci-	d	$\times 10^{-1}$
centi-	c	$\times 10^{-2}$
milli-	m	$\times 10^{-3}$
micro-	µ	$\times 10^{-6}$
nano-	n	$\times 10^{-9}$
pico-	p	$\times 10^{-12}$
femto-	f	$\times 10^{-15}$
atto-	a	$\times 10^{-18}$

- **Derived units** are created by multiplication and/or division of the base units
  - Speed: $m \cdot s^{-1}$ or m/s
  - Volume: 1 cm^3 = 1 mL = 10^{-3} L
  - Density: g/cm^3

- Note that dividing by a unit is mathematically equivalent to multiplying by that unit to the –1 power

- If a derived unit has a common name, it is typically represented with a capital letter
  - Force: 1 Newton (1 N) = 1 $kg \cdot m \cdot s^{-2}$
  - Heat, work, or energy: 1 Joule (1 J) = 1 $kg \cdot m^2 \cdot s^{-2}$
  - Pressure: 1 Pascal (1 Pa) = 1 $N \cdot m^{-2}$ = 1 $kg \cdot m^{-1} \cdot s^{-2}$

- Scientific measurements are subject to human limitations on accuracy and precision.
- **Accuracy** refers to how close a measurement (or the mean of a set of measurements) is to an actual or “true” value.
- **Precision** refers to how close a series of measurements are to one another or how reproducible they are.

(a) **Precise, but not accurate**

(b) **Both precise and accurate**

(c) **Neither precise nor accurate**

Figure 1.13

- Scientific measurements are reported so that every digit is certain *except* the last, which is estimated.
- The number of digits reported in a measurement depends on the measuring instrument or device.
- For analogue measurements (on a visual scale): always estimate to one digit beyond the smallest scale division, if possible.

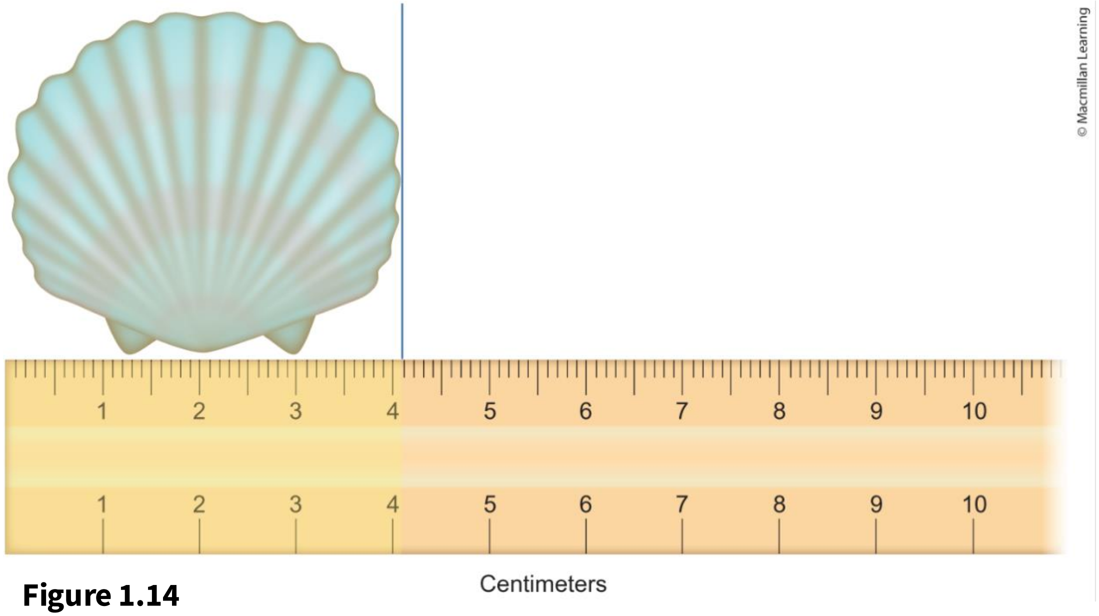

Figure 1.14

- **Significant digits (figures)** are non-placeholding digits that communicate the value of a measurement as opposed to its power of ten.
- Equivalently, significant digits are those that appear in the **significand** when the number is written in scientific notation (n in $n \times 10^m$).
- The greater the number of significant digits, the greater the certainty of the measurement.

$$\mathbf{1.0555} \times 10^2 \text{ g}$$

- **Rules for determining significant digits:**
  1. All nonzero digits are significant.
  2. Zeros between two significant figures are themselves significant.
  3. Zeros at the beginning of a number (leading zeros) are never significant.
  4. Zeros at the end of a number are significant and after the decimal point are always significant.
  5. Zeros at the end of a number and before the decimal point are not significant if a decimal point is not present.

- Note that **exact numbers**, including counting numbers and mathematical constants (π, e), do not limit significant digits in calculations involving these numbers

1. **For addition and subtraction.** The result has the same level of precision as the least precise measurement. In other words, the number of decimal places in the result is the same as the measurement with the smallest number of decimal places.

   Example: Adding or subtracting two volumes

$$\begin{array}{r} 865.\mathbf{9}\ \text{mL} \\ -\ 2.8121\ \text{mL} \\ \hline 863.0879\ \text{mL} \end{array} = \mathbf{863.1}\ \text{mL}$$

$$\begin{array}{r} 83.\mathbf{5}\ \text{mL} \\ +\ 23.28\ \text{mL} \\ \hline 106.78\ \text{mL} \end{array} = \mathbf{106.8}\ \text{mL}$$

2. **For multiplication and division.** The result has the same number of significant digits as the least precise measurement. In other words, the least precise measurement limits the precision of the result.

   Example: Calculating a volume

$$\mathbf{9.2}\ \text{cm} \times 6.8\ \text{cm} \times 0.3744\ \text{cm} = 23.4225\ \text{cm}^3 = \mathbf{23}\ \text{cm}^3$$

1. **Rounding with the first digit removed *greater than 5.*** The least significant digit in the result is increased by 1.

   Example: 5.379 rounds to 5.38 or 5.4.

2. **Rounding with the first digit removed *less than 5.*** The least significant digit in the result is unchanged.

   Example: 0.2413 rounds to 0.241 or 0.24.

3. **Rounding with the *only* digit removed *equal to 5.*** Round up if the preceding digit is odd and round down if the preceding digit is even.

4. **Rounding with the first digit removed *equal than 5.***
   Round up (rule 1) unless there are subsequent digits above zero.

As a rule, don't worry about rounding or significant digits until the very end of a calculation.
This is not something you should be stressing about in an introductory chemistry course.

1.7 Dimensional Analysis

- Many algorithmic problems in CHEM 1211K and 1212K (not all!) will involve linearly related measurements. These can (and should) be tackled using a systematic approach

- **State the Problem.** What is the problem asking for? Include units and, if applicable, a level of precision.

- **Plan.**
    1. Clarify the **known** and **unknown quantities**, including units
    2. Determine steps to work from the known(s) to the unknown using **conversion factors**
    3. Create a visual summary of the steps involved

- **Solve.** Use **dimensional analysis** to set up a series of conversion factors to arrive at the answer and calculate.

- **Check.** Do the value and units of the answer make sense? Is the value on the right scale? Do the units correspond to the phenomenon or quantity asked for?

- We use **dimensional analysis** to convert one quantity into another related to the first in a linear way
- The quantities involved may be the same phenomenon expressed in different units (centimeters, inches) or different phenomena related linearly (Joules, degrees Celsius)
- **Conversion factors** are ratios based on expressing the same quantity on two different unit scales; we use these in dimensional analysis to convert between units

$$2.54 \text{ cm} = 1 \text{ in}$$

$$\frac{2.54 \text{ cm}}{1 \text{ in}} = 2.54 \text{ cm/in}$$

$$\frac{1 \text{ in}}{2.54 \text{ cm}} = 0.394 \text{ in/cm}$$

- Set up the conversion factor to divide out the unit of the known (in the denominator) and leave behind the unit of the unknown (numerator)

$$\text{given unit}\left(\frac{\text{desired unit}}{\text{given unit}}\right) = \text{desired unit}$$

- **Example.** Convert 8.00 meters (8.00 m) to inches (in), given that 2.54 cm is equivalent to 1 in.
  1. First, convert meters to centimeters.
  2. Then, convert centimeters to inches.

$$8.00\ \text{m}\left(\frac{10^2\ \text{cm}}{1\ \text{m}}\right)\left(\frac{1\ \text{in}}{2.54\ \text{cm}}\right) = 315\ \text{in}$$

1.9 Temperature Scales

- In scientific contexts, the **degrees Celsius (°C)** and **Kelvin (K)** scales are most used for temperature

- The Celsius scale is based on the properties of water: water freezes at 0 °C and boils at 100 °C

- The Kelvin scale is based on the absolute temperature and kinetic energy of an ideal gas: at 0 K, all molecules are unmoving. Water freezes at 273.15 K and boils at 373.15 K

- Note that Kelvin temperatures cannot be negative and that –273.15 °C is the minimum possible value for the Celsius temperature scale (sanity checks!)

$$\mathrm{K} = \ {}^{\circ}\mathrm{C} + 273.15$$

Need to Review?

- Check out the Fundamentals of Chemistry open course on Canvas for additional review resources. Click the link to access the course!

- Prof. Geoff Hutchinson at the University of Pittsburgh has put together a wonderful Mathematics for Chemistry course on Github. See lectures 0 – 2 and 9 for math relevant to CHEM 1211K and 1212K.